

Full Project Consultation Transcript

Topic: Automated Corn Phenotyping via 3D-to-2D Synthetic Projections

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USER

I think for baseline, getting euclidean distance serves better

AI

Using Euclidean distance as a baseline is a scientifically sound choice because it represents a 'Geometric Baseline.' I suggested Option A (Naive 2D) to quantify perspective distortion and Option B (Oracle 3D) to show the intrinsic error of ignoring stem tortuosity. Proposed metrics included Percent Error of Internode Length and Error Projection Overlay.

USER

I was thinking of using resnet based baseline but for regression(input of image and its node lengths associated with the keypoints), which pretrained model do you suggest?

AI

Suggested ResNet-18 for its speed and lower risk of overfitting on synthetic data, or ResNet-50 for deeper feature extraction. Discussed architectural adjustments like replacing the FC layer, using Global Average Pooling, and selecting MSE or Huber Loss.

USER

I ditched Isaac sim... created intrinsics of a virtual camera, multiple extrinsics and projected the images and got the 2D dataset. I also got the keypoints from the ply file as ground truth.

AI

Critiqued 'Point' vs 'Texture' rendering (suggested splatting). Offered Path 1 (Implicit Regressor: Image to Lengths) and Path 2 (Explicit Keypoint Regressor: Image to Coordinates). Suggested ResNet-34 as a 'sweet spot' and discussed View-Invariance metrics.

USER

I am doing the path one

AI

Provided PyTorch code for a ResNet-18 regression head with BatchNorm and Dropout. Emphasized target normalization (scaling to [0,1]) and the use of Mean Absolute Percentage Error (MAPE) for agricultural phenotyping.

USER

I used three pipelines: resnet 34, efficient net and custom CNN with no pretrained weights. [Provided Codebase]

AI

Analyzed the provided code. Highlighted the importance of the ``masked_mse`` loss. Proposed visualization options: Data Flow (Option 1), Geometric Priors (Option 2), and Loss Function/Masking (Option 3).

USER

Write this MLP head in equation form [Code provided]

AI

Formatted the MLP head as a composite function: $y_{\text{hat}} = \sigma(W3 \cdot \sigma(\text{BN2}(W2 \cdot \sigma(\text{BN1}(W1x + b1)) + b2)) + b3)$. Provided concise linear formats for use in Microsoft Word.

USER

Write mathematical equation for resnet34 too. make it in linear format.

AI

Defined ResNet-34 using the Residual Function ($z_{out} = \sigma(F(z, \{W\}) + W_s \cdot z)$), Encoder Feature Extraction ($g = P(E(x))$), and the Regression Mapping ($y_{hat} = H(g)$).

USER

Similarly for efficientnet_b3 too.

AI

Defined EfficientNet-B3 using Mobile Inverted Bottleneck (MBConv) logic: $z_{out} = SE(\text{Depthwise_Conv}(\text{Inverted_Bottleneck}(z)))$ and the final 1536-dimensional regression mapping.

USER

Now for my untrained model [CustomCNNReg] and masked_mse loss.

AI

Formalized CustomCNNReg (8 residual blocks) and the Masked Mean Squared Error: $L = \Sigma (y_{hat} - y_{true})^2 \cdot m) / \Sigma (m)$.

USER

Explain wilcoxon rank high test for this context

AI

Explained the Wilcoxon Signed-Rank Test for paired comparison. Defined the null hypothesis ($H_0: \text{Med}(d_i) \leq 0$) and the procedure for ranking differences in Absolute Error between models.

USER

Act as a strict academic editor... [Governance & Rules provided]

AI

Accepted persona. Initiated reverse-prompting protocol for the Introduction section based on MaizeField3D dataset context and 75x annotation efficiency gain.

USER

Act as a presentation coach... [Slide deck requirements provided]

AI

Developed a 15-slide deck outline including mandatory HW12 reflection slides, APA-style citations, and technical walkthroughs using inline code.

USER

Generative AI Governance: Augment capabilities, not replace... Declaration of compute limits... Lead architect vs Junior assistant.

AI

Incorporated governance constraints into a Consultation Log & AI Audit Kit (PDF v2). Verified Lead Architect status.